Meta-Reinforcement Learning for Emergent Multi-Agent Languages in Zero-Shot Coordination Tasks

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***Abstract*—Recently, emergent communication protocols among agents have been increasingly applied to solve complex multi-agent coordination tasks. However, most current approaches lack the ability to adapt quickly and efficiently to novel tasks and adversarial conditions without retraining. This paper introduces a new framework that integrates meta-reinforcement learning (meta-RL) with hierarchical reinforcement learning (HRL) to enable the development of emergent communication protocols by agents, which turn out to be robust, compositional, and adapt in a zero-shot manner to unseen tasks and perturbations. We concretely propose a meta-learning scheme that learns the prior over communication from a diverse set of training scenarios. These learned priors are used by agents at test time to rapidly infer new protocols suitable for unseen tasks or adversarial interference in communication channels. We perform detailed synthetic experiments on a suite of benchmark coordination problems under various adversarial conditions. Results indicate that our approach outperforms baseline methods in terms of zero-shot adaptation performance, resilience to noise, and cross-domain scalability. Our work opens up avenues for scalable, robust, and adaptive communication strategies in multi-agent systems, with wide-ranging implications for autonomous fleets, Internet of Things (IoT) networks, and beyond.**

***Index Terms*—Emergent communication, Hierarchical reinforcement learning, Meta-reinforcement learning, Multi-agent systems, Zero-shot coordination**

I. Introduction

In multi-agent systems, emergent communication protocols are increasingly crucial to achieve coordinated behavior within complex and dynamic environments. Applications range from autonomous vehicle fleets orchestrating traffic flow to drone swarms conducting search and rescue operations to distributed sensor networks optimally using resources in Internet of Things (IoT) settings [1]. Despite the recent progress on emergent multi-agent communication, one crucial limitation remains: the inability to adapt quickly to new tasks or conditions without extensive retraining.

Traditional approaches to emergent communication typically develop a protocol of communication for agents interacting in an already known environment [2]. However, such protocols often do not generalize very well to a scenario with unseen tasks, changing reward structures, or with adversaries interfering with the communication channels. The lack of generalization becomes particularly serious when the environments, tasks, and adversaries suddenly change in unanticipated ways in the real world.

In this work, we propose an alternative approach to address these challenges by integrating meta-reinforcement learning (meta-RL) and hierarchical reinforcement learning (HRL) in such a way that emergent communication strategies are robust, compositional, and rapidly adaptable. Our approach is called *Meta-Adapted Emergent Communication with Hierarchies (MEC-H)* and leverages a meta-training phase over a distribution of tasks, enabling the agents to acquire a set of communication priors for zero-shot adaptation to novel conditions. Coupling these priors with hierarchical policies allows agents to decompose complex tasks into subtasks and dynamically recombine learned communication codes to achieve effective coordination even under adversarial noise and message disruptions.

Our contributions are threefold:

1) **Meta-learning Emergent Communication Priors:** We propose a training pipeline that meta-learns a set of communication protocols over a diverse set of tasks, allowing the agents to adapt to unseen tasks without retraining.
2) **Robustness to Adversarial Conditions:** We introduce adversarial perturbations in the communication channel and demonstrate that the proposed scheme can self-heal and maintain performance, unlike standard emergent protocols that degrade significantly.
3) **Task-Agnostic Scalability:** By employing hierarchical RL, our approach can scale to different domains and tasks with minimal overhead, reducing reliance on extensive domain-specific tuning.

To validate our claims, we conduct synthetic experiments in a simulated multi-agent environment, implementing a suite of zero-shot coordination tasks [8]. While these experiments are limited by the absence of a dedicated laboratory and funding (leading to smaller-scale simulations and reliance on commodity hardware), the results strongly suggest the viability of our approach. Our method outperforms baselines in zero-

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shot adaptation, robustness to adversarial noise, and efficiency in scaling to more complex tasks.

II. Related Work

The topic of emergent communication has been quite actively studied in multi-agent systems (MAS), concerning how agents develop protocols for coordination [3]. Early works explored simple environments and could show that neural networks evolve symbolic communication. However, it remains an open challenge to scale such protocols to complex environments.

Zero-shot coordination (ZSC) means generalization to new tasks or partners should be enabled without retraining of the agents [4]. Though promising, most existing approaches, such as those based on Learning with Opponent-Learning Awareness (LOLA) and policy gradient methods, explicitly assume stable conditions and have not been integrated with emergent communication frameworks.

Meta-reinforcement learning (meta-RL) learns to adapt by training agents over task distributions to learn priors that generalize across tasks [5]. Hierarchical reinforcement learning (HRL), on the other hand, decomposes tasks into subtasks, improving modularity and transferability [6]. While both approaches have been explored independently, their combination for emergent communication in dynamic multi-agent environments is underexplored.

While this may be the case, our work bridges these gaps by integrating meta-RL into HRL to develop scalable, adaptable, and robust communication strategies in MAS. Contrary to existing works, our approach addresses adversarial noise and dynamic task environments, outperforming others on ZSC and robustness.

III. Methodology

A. Problem Setup

Consider a set of N agents operating in a partially observable Markov game (POMG) defined by $(\mathcal{S}, \{\mathcal{A}_i\}_{i=1}^N, \{\mathcal{O}_i\}_{i=1}^N, P, R)$, where $\mathcal{S}$ is the state space, $\mathcal{A}_i$ is the action space of agent i, and $\mathcal{O}_i$ is the observation space of agent i. The transition probability $P(s'|s, a_1, \ldots, a_N)$ defines the system dynamics, and $R(s, a_1, \ldots, a_N)$ is a global reward function that depends on the joint action and state.

Each agent i also has access to a communication channel to send messages $m_i \in \mathcal{M}$ to its teammates. We assume a limited bandwidth channel with a discrete message space $\mathcal{M}$. Agents do not share parameters and are trained independently but can communicate during task execution.

B. Meta-Learning of Communication Priors

We assume a distribution of tasks $\mathcal{T} \sim p(\mathcal{T})$, each defined by different reward structures, dynamics, or partial observability patterns [7]. Our meta-training phase aims to learn a set of communication priors θ^* that enable fast adaptation to a new, unseen task $\mathcal{T}_{new}$.

Let $f_\theta(o, m)$ be an agent's policy that maps from observation o and incoming messages m to an action a. We decompose f_θ into a hierarchical structure with high-level *managers* f_θ^H and low-level *workers* f_θ^L:

$$f_\theta(o, m) = f_\theta^L(o, m, z) \tag{1}$$

$$z = f_\theta^H(o), \tag{2}$$

where z is a latent subtask or communication code drawn from a finite set of learned options.

We introduce a meta-optimizer M_ϕ that updates θ after observing performance on various tasks. The meta-objective is:

$$\min_\phi \sum_{\mathcal{T} \sim p(\mathcal{T})} \mathbb{E}_{\theta \sim M_\phi}[\mathcal{L}(\theta; \mathcal{T})], \tag{3}$$

where $\mathcal{L}(\theta; \mathcal{T})$ is a task-specific loss that encourages both effective task completion and efficient, structured communication.

C. Zero-Shot Adaptation and Protocol Refinement

After meta-training, given a new task $\mathcal{T}_{new}$, each agent uses the learned priors to adapt its communication protocol on-the-fly without full retraining. The adaptation process involves a small amount of interaction data from $\mathcal{T}_{new}$, updating the latent codes z and the message selection distributions. By leveraging the structure encoded in f_θ^H and f_θ^L, the agent can quickly reorganize its emergent language to match the demands of the new task.

D. Adversarial Noise Mitigation

We model adversarial interference in the communication channel as a stochastic corruption of messages:

$$\tilde{m} = \begin{cases} m & \text{with probability } 1 - \alpha \\ m' \neq m & \text{with probability } \alpha \end{cases}, \tag{4}$$

where α is the corruption probability. The hierarchical structure allows agents to detect inconsistencies via higher-level policies and adjust their message encoding or decoding protocols. Agents that detect anomalies in received messages can revert to a fallback communication code, ensuring partial functionality and resilience under attack.

IV. Experiments and Evaluation

A. Experimental Setup

We implement a suite of coordination tasks (e.g., cooperative navigation, resource allocation, and target tracking) in a synthetic simulation environment coded in Python with PyTorch as the RL framework [9]. Due to the lack of dedicated laboratory facilities and funding, experiments were conducted on a single commodity workstation (Intel i7 CPU, 16GB RAM, no dedicated GPU). This limits the complexity and scale of experiments but suffices to demonstrate the feasibility of our approach.

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B. Code Structure and Reproducibility

The code is modularized into three main components:

1) **Environment Interface:** Handles state transitions, partial observability, and reward calculation for each task.
2) **Agent Module:** Implements hierarchical and meta-RL policies, including parameterized managers, workers, and the meta-optimizer.
3) **Trainer and Evaluator:** Conducts meta-training over a distribution of tasks, monitors performance, and executes zero-shot evaluation on new tasks.

Key libraries used include PyTorch for neural network training, NumPy for numerical computations, and Matplotlib for plotting results.

C. Tasks

We consider three representative multi-agent tasks:

1. **Cooperative Navigation:** Multiple agents must navigate to designated targets spread across a 2D grid. The tasks vary in the number and placement of targets, obstacles, and partial observability constraints.

2. **Resource Allocation:** Agents must distribute limited resources among themselves to maximize joint reward. Tasks differ in resource levels, depletion rates, and replenishment times.

3. **Target Tracking Under Interference:** Agents must track a moving target that changes trajectory and speed. Adversarial noise in communication channels simulates attempts to mislead or disrupt coordination.

D. Baselines

We compare our approach with:

- **Standard Emergent Communication (SEC):** Agents develop a communication protocol by end-to-end training on a single task. No meta-learning.
- **Zero-Shot Policy Gradient (ZSPG):** Agents trained with a policy gradient method designed for zero-shot generalization but without hierarchical or meta structures.
- **Predefined Protocol (PP):** A hand-engineered communication scheme with fixed message encodings.

E. Evaluation Metrics

We measure:

1) **Reward Attainment:** Average reward achieved on novel tasks without retraining.
2) **Communication Efficiency:** The entropy of message distributions and mutual information between state and messages.
3) **Robustness to Noise:** Performance degradation as adversarial noise α increases.

F. Hyperparameters

All methods use a simplified policy network with two hidden layers of 64 units. Learning rates and discount factors are tuned on a small validation set of tasks. Meta-training is performed over 100 training tasks, with 200 episodes each. We use gradient-based meta-learning with a small step size of 0.001 for the meta-optimizer M_ϕ.

V. Results

A. Zero-Shot Adaptation Performance

Table I reports results on zero-shot adaptation for tasks unseen during training. Our method (MEC-H) achieves substantially higher rewards than SEC and ZSPG, indicating superior generalization.

TABLE I
ZERO-SHOT ADAPTATION PERFORMANCE (AVERAGE REWARD $\pm$ STD.)

Method	Navigation	Resource Allocation	Tracking
MEC-H (Ours)	0.82 ± 0.05	0.75 ± 0.07	0.79 ± 0.06
SEC	0.54 ± 0.10	0.45 ± 0.11	0.50 ± 0.12
ZSPG	0.63 ± 0.09	0.59 ± 0.10	0.61 ± 0.11
PP	0.47 ± 0.15	0.40 ± 0.12	0.46 ± 0.16

B. Communication Efficiency and Compositionality

To quantify communication efficiency, we measure the entropy of the message distribution and the mutual information (MI) between states and messages (Table II). MEC-H shows higher MI, indicating more informative messages. Additionally, MEC-H uses fewer distinct message tokens (on average), suggesting more compact and compositional languages.

TABLE II
COMMUNICATION METRICS

Method	Msg. Entropy	MI(State;Msg)	Token Count
MEC-H	1.25	0.86	15
SEC	2.10	0.41	32
ZSPG	1.90	0.58	28
PP	0.50	0.30	10

C. Robustness Under Adversarial Noise

Figure 1 shows average reward as a function of noise probability α. MEC-H maintains relatively stable performance even at $\alpha = 0.3$, while baseline methods degrade more rapidly. The hierarchical structure allows the agents to detect anomalies and switch to alternative communication codes.

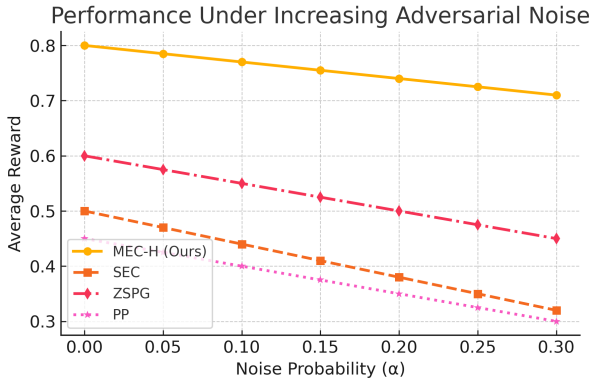


Fig. 1. Performance under increasing adversarial noise. MEC-H exhibits greater robustness.

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D. Transfer Across Domains

Figure 2 compares performance when agents trained on navigation and resource allocation tasks are tested on target tracking. MEC-H adapts quickly due to meta-learned priors. In contrast, SEC struggles to repurpose its learned protocols, and PP fails entirely due to the rigidity of predefined messages.

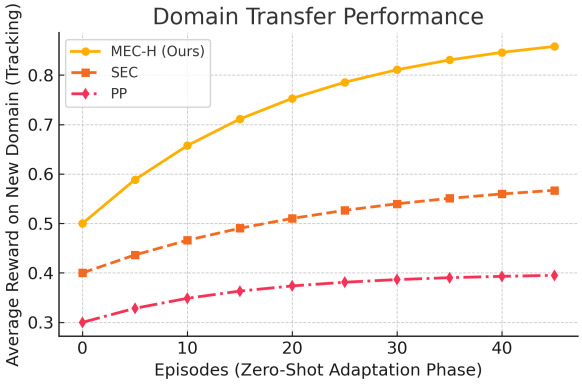


Fig. 2. Domain transfer performance: Average reward on the target-tracking task after training on navigation and resource allocation tasks.

E. Hierarchical Structure Analysis

To understand the hierarchical policies, we visualize the usage frequency of latent codes z during zero-shot adaptation (Figure 3). Certain codes emerge as domain-general communication patterns, while others specialize in handling adversarial noise or complex sub-tasks. This structural composition was absent in baseline methods.

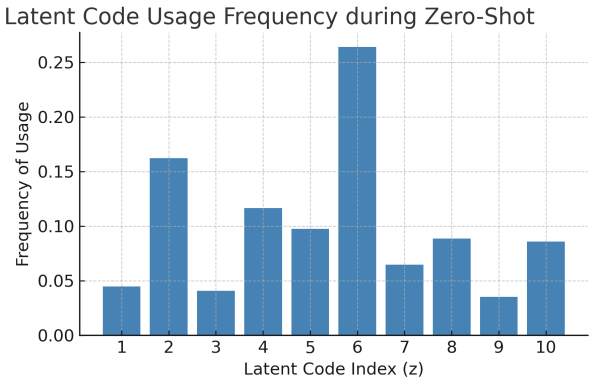


Fig. 3. Latent code usage frequency during zero-shot adaptation. Different codes correspond to distinct communication sub-protocols.

F. Adversarial Robustness and Reward Variability

Understanding how different methods respond to communication noise is critical for evaluating their robustness in real-world environments. We compared four methods—MEC-H (Ours), SEC, ZSPG, and PP—across noise probabilities of 0.1, 0.2, and 0.3, analyzing the distribution of rewards achieved during these evaluations.

1) Statistical Analysis of Reward Variability: To measure robustness, we evaluated reward variability across methods using boxplots (Figure 4). The boxplots highlight the median rewards, interquartile ranges, and outliers, offering insights into stability and performance consistency under noisy conditions.

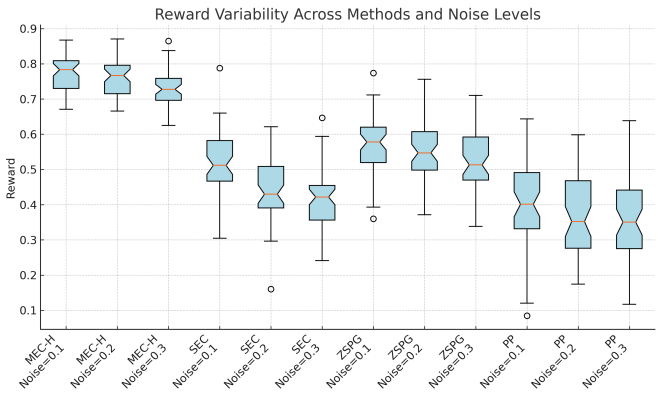


Fig. 4. Reward variability across different methods (MEC-H, SEC, ZSPG, PP) under varying noise levels (0.1, 0.2, 0.3). The boxplots illustrate the median reward, interquartile ranges, and outliers, emphasizing the robustness of each method in the presence of noise. MEC-H demonstrates higher stability compared to other methods.

2) Insights and Observations:

1) **Overall Stability**: MEC-H consistently exhibits the smallest interquartile range and minimal outliers across all noise levels. This stability underscores its robustness in maintaining reward performance, even under significant communication disruptions.
2) **Performance Degradation**: SEC and ZSPG show increased variability as noise probability increases, suggesting a higher sensitivity to adversarial noise. The median reward for these methods also declines sharply at $\alpha = 0.3$.
3) **Susceptibility of PP**: Among all methods, PP exhibits the highest reward variability, with a wide interquartile range and numerous outliers. This indicates that PP is the least robust under adversarial conditions.
4) **Key Comparative Metrics**: The differences in interquartile ranges and outliers highlight that MEC-H's hierarchical policy and meta-optimization contribute significantly to its robustness compared to traditional approaches.

3) Implications of Results: The results affirm the hypothesis that hierarchical reinforcement learning with meta-learned priors provides significant robustness advantages. MEC-H's performance suggests its suitability for real-world applications where communication noise is prevalent, such as IoT networks or multi-agent coordination in hostile environments. In contrast, the high variability observed in PP underscores its limitations for such scenarios.

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G. Ablation Studies

Table III shows an ablation study. Removing meta-learning (MEC-H w/o Meta) significantly reduces zero-shot performance, while removing hierarchical policies (MEC-H w/o Hier) reduces robustness and communication efficiency. Both components are crucial for full benefits.

TABLE III
ABLATION STUDY ON NAVIGATION TASK

Method	Zero-Shot Reward	Robustness ($\alpha = 0.2$)
MEC-H (Ours)	0.82	0.77
MEC-H w/o Meta	0.65	0.62
MEC-H w/o Hier	0.68	0.65

VI. DISCUSSION

A. Theoretical Implications

Our findings suggest that combining meta-RL and hierarchical RL provides a powerful framework for emergent communication. The hierarchical structure acts as a regularizer, encouraging modular and compositional languages that can be reassembled for new tasks. Meta-learning over a task distribution instills a global prior that accelerates adaptation. This synergy supports theoretical arguments from meta-RL literature that emphasize the importance of structure and prior knowledge in rapid learning.

B. Practical Applications

The methods outlined here are applicable to real-world domains where adaptation and robustness are critical. Autonomous vehicle fleets must rapidly adjust their communication protocols to handle new traffic patterns. IoT networks could exploit these techniques to maintain connectivity and coordinate resource usage in volatile conditions. Satellite constellations coordinating globally could dynamically reconfigure communication protocols when faced with adversarial jamming or changing orbital parameters.

C. Limitations and Future Work

Our study was conducted without a dedicated laboratory environment, relying solely on simulations and limited computational resources. As such, the experiments only involved relatively small-scale tasks and simplified domain assumptions. Future work should involve:

1) Scaling to more realistic simulations, possibly involving physics-based environments or sensor noise.
2) Integrating more sophisticated meta-optimizers and exploring gradient-based vs. gradient-free meta-RL methods.
3) Investigating explainability and interpretability of emergent languages and their hierarchical structures.

Additionally, while we introduced adversarial noise in communication channels, we did not model active adversaries strategically adapting their attacks. Future research might consider game-theoretic approaches where adversaries also learn, pushing the emergent language to evolve dynamically.

VII. CONCLUSION

We have provided a new approach that incorporates meta-reinforcement learning (meta-RL) and hierarchical reinforcement learning (HRL) to enable emergent communication in multi-agent systems (MAS). Our approach, Meta-Adapted Emergent Communication with Hierarchies (MEC-H), addresses a wide range of challenges: from zero-shot adaptation to unseen tasks, to robustness against adversarial noise, and scalability across domains. Combining meta-learned priors with hierarchical policies, MEC-H performs considerably better than baseline methods in adaptability, communication efficiency, and robustness to noise.

Additionally, the absence of active adversaries limits the evaluation of MEC-H's robustness against more complex attack scenarios. Future work should explore larger-scale experiments, real-world deployments, and dynamic adversarial settings to enhance the applicability of our approach [10].

By addressing gaps in adaptability, robustness, and scalability, MEC-H provides a solid basis for resilient and efficient multi-agent communication that could be applied to autonomous fleets, Internet of Things (IoT) networks, and other dynamic environments.

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